

KrishiRover: An Autonomous Farming Rover for Soil Monitoring, Smart Irrigation, and Bangla Voice Interaction

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Abstract—In the resource-scarce places like Bangladesh, it is required to have some technologies that can help us with our manual labor, irrigation decision making, and communication of the agricultural data in a friendly manner. For that purpose, we have proposed a very affordable four-wheel autonomous agricultural robot, called KrishiRover that can perform the tasks like movement, soil moisture sensing, temperature sensing, humidity sensing, obstacle detection and avoidance, automatic watering, IoT-enabled monitoring, and Bangla voice communication. The rover can be controlled by an ESP32 and includes ultrasonic, soil moisture, and DHT11 sensors, motor driver, relay-based water pump, and Bangla voice system. The rover is able to maintain the field path, re-check soil status in various zones, provide automatic water for the plants, and communicate its status through a web interface. This paper describes the system architecture, hardware software integration, process flow, and testing of the process. The paper demonstrates how it is possible to develop autonomous navigation, intelligent irrigation, IoT enabled monitoring, and native language communication on a simple and affordable platform.

Index Terms—agricultural robotics, smart irrigation, soil moisture monitoring, Internet of Things, ESP32, obstacle avoidance, autonomous rover, Bangla voice assistant, precision agriculture, Bangladesh

I. INTRODUCTION

Agriculture is becoming increasingly dependent on data collection that allows for monitoring of conditions in fields, the needs of crops, the availability of water, and the condition of farm equipment. Precision farming supported by IoT links sensors, communication tools, processing systems, and actuators and enables better decisions based on up-to-date data rather than guesswork and field visits only [1], [2], [3]. Yet while the availability of cheap sensors, wireless communications, cloud services, and AI applications are lowering the threshold of entry into precision agriculture, the high cost of implementation, poor rural infrastructure, the accuracy and

stability of sensors, and the lack of technical skills are major obstacles for its adoption [4], [5], [6], [7]. Water management is another crucial element of intelligent agriculture because time-based irrigation can supply either excess or insufficient amount of water, having a negative effect on the plant growth process. Water content in soil, temperature, humidity, rainfall, and the state of crops are factors that have to be considered when determining the starting and finishing times of irrigation and are widely used as an input in automatic irrigation systems [8], [9], [10]. The performance of this system depends on agricultural environment, sensor accuracy, plant properties, reliability of communication channels, and control approach chosen [11], [12], [13]. Mobile robots in agriculture are capable of gathering data from various parts of the field and act based on spatial information. Previous works have focused on the use of robots for soil analysis, crop analysis, field surveying, and automated irrigation, using technologies such as LiDAR, cameras, robotic arms, and computing units [14], [15], [16], [17], [18]. These technologies are technically sophisticated but too expensive and complicated for use in small farms or for academic prototypes. KrishiRover proposes an inexpensive solution using a four-wheel base combined with three ultrasonic sensors, soil and environmental sensors, automated irrigation, IoT dashboard, and Bangla voice output. The robot shows simple facilities for autonomous navigation, avoiding obstacles, area-based soil monitoring, targeted irrigation, and human-robot interaction.

II. LITERATURE REVIEW AND RELATED WORK

A. Agricultural IoT and Smart-Farming Platforms

IoT systems used in farming typically involve the use of many layers, including the sensing, communication, processing, and application. Sensors are responsible for data collection, while communication technologies, such as Wi-

Fi, Zigbee, LoRa, or cellular networks transmit this data, and cloud or edge platforms process the information before presenting it to users or using it for controlling actuators (Quy, 2022; Xu, 2022). Latest research shows that smart-farming platforms are increasingly using combinations of IoT devices with sensors, mobile applications, cloud services, robotics, and artificial intelligence. However, it is important for the technology architecture to be in accordance with the farm size, availability of power, communication distance, maintenance skills and users' technical capabilities (Qazi & Tokunaga, 2022; Miller et al., 2022). IoT-based irrigation usually involves the use of ESP32, Arduino, NodeMCU, or Raspberry Pi with soil-moisture and environmental sensors, which usually send data to a dashboard and control pumps and valves according to the sensor information (Obaideen et al., 2022; Morchid et al., 2024; Huque & Chaudhuri, 2023). KrishiRover employs the same principles of sensing, communication, and action, but applies them to a rover that implements sensing and irrigation functions.

B. Soil and Environmental Sensing

Effective irrigation depends on precise measurements of the soil and environment parameters. There are several methods to measure the soil moisture, which include resistive, capacitive, dielectric, time-domain reflectometry, electromagnetic and remote sensing methods, with varied costs, accuracies and complexities [19], [20]. Cost-effective capacitive sensors are commonly used for small irrigation schemes but their readings might depend on soil type, depth of insertion, corrosion and calibration accuracy [11]. The fixed sensor network delivers continuous readings from selected points, whereas a mobile device can monitor several areas with fewer sensors [21], [22]. Humidity and temperature are also important as they affect evaporation, plants' state and the soil moisture reading interpretation [23]. Wide-area monitoring can be done using satellite, UAV, LiDAR, radar, and multispectral image analysis [24], [25]. Saha et al. created a Bangladesh-based precision agriculture framework with satellite analysis, soil sensing, crops' recommendations and fuzzy irrigation [26]. The soil-moisture sensor is the primary sensor in KrishiRover irrigation, while DHT11 sensor delivers additional temperature and humidity data.

C. Agricultural Rovers and Obstacle Avoidance

Agricultural robots are required to navigate through rough terrain and sparsely present environments with safety. High-tech systems typically apply cameras and LiDAR to figure out patterns, detection of obstacles, and figure out safe paths to take [27]. Obstacle avoidance methods may be based on the reactivity of the work, creating maps, using machine learning, sensor-based methods, or hybrid methods, depending on computational power and available sensors [28]. These methods can provide better perception, but at the same time, they add value to the price and complexity of the system. Various robots have been invented for monitoring crops and soil. The autonomous platforms are used in soil resistivity mapping, volumetric water content measurements, cropping images, insertion of the probe, and watering the plants [14], [15],

[16], [17], [18]. KrishiRover approaches the task differently as it is cheaper and easier than other methods. In particular, it uses three HC-SR04 ultrasonics to measure distances from the rover to obstacles without LiDAR, cameras, or robotic arms being involved in the process.

D. Smart Irrigation and Decision Support

Inexpensive methods of irrigation commonly apply easy moisture thresholds for pump control. Such systems are rather simple in application and operation, especially in small agricultural applications. Recently developed systems based on the ESP32 application use features like soil moisture detection, monitoring via the sensors, opening and closing of valves, and Internet of Things (IoT) applications [29]. KrishiRover has a similar functioning, the only modification is that KrishiRover adds charging movement, obstacle avoidance and Bangla voice interaction to the same system. Among the advanced options for irrigation management are machine learning, fuzzy logic, reinforcement learning, and digital twins [10], [30], [31], [32]. These options appear to be good in decision-making, but require some reliable data, processing effects, and proper validation [33], [34]. Given the fact that the results differ with the soils, crops and climates [35], [36], KrishiRover now opts for ruling based method of irrigation.

E. Wireless Communication and User Interaction

Communication in smart agriculture relies on the following parameters including field dimensions, power consumption, data rates, and infrastructure requirements. While WiFi allows for a higher data rate of transmission with a relatively short-range coverage, Zigbee offers low-power communications. LoRaWAN, Sigfox, and NB-IoT offer broader coverage yet vary in terms of message sizes, latency, infrastructure, and operating costs [4], [37]. Studies of large fields have revealed that LPWAN allows for better coverage than Zigbee, particularly for near-ground sensors in hilly regions [37]. The KrishiRover relies on WiFi since it works within a defined experimental field and transmits small JSON messages to a local server. These messages contain soil condition, temperature, humidity, pump state, obstacle distance, zone, orientation, and rover state. To increase accessibility, KrishiRover uses a Bangla assistant called SechBondhu which answers user's questions based on the latest data collected by rover and does not generate any sensor value when current data is not available.

F. Comparative Analysis of Selected Systems

The reviewed studies address a number of interconnected areas of smart farming, such as Internet of Things (IoT), irrigation, robots, sensing, artificial intelligence (AI), and communication. While most studies involve stationary monitoring or sophisticated research facilities, the KrishiRover integrates all these technologies into an easier mobile prototype.

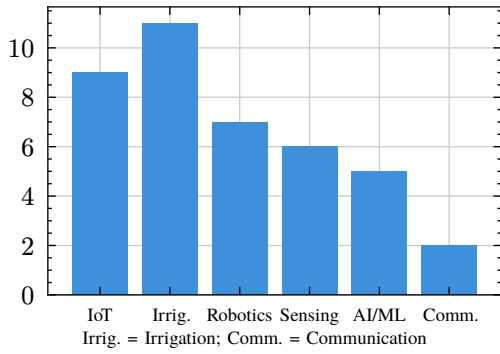


Fig. 1. Dominant focus distribution within the reviewed literature corpus.

The following graph presents the dominant technical focus of the literature used in this report. These values represent a report-level classification of the reviewed papers rather than experimental performance.

TABLE I

COMPARISON OF REPRESENTATIVE RELATED SYSTEMS WITH THE PROPOSED KRISHIROVER PLATFORM.

Study	Focus	Technologies	Limitation
Kumar et al. [8]	Irrigation	ESP32, soil/climate sensors, cloud	Fixed system; no rover navigation
Han et al. [27]	Navigation	LiDAR, camera, vision	Higher cost and computation
Unal et al. [14]	Soil mapping	Autonomous rover, DGPS, Wenner probes	Large specialized platform
Linford and Haghshenas-Jaryani [15]	Crop/soil monitoring	4WD rover, 6-DOF arm, camera, RTK	Complex hardware and localization
Tsimpidi et al. [16]	Moisture sensing	UGV, TEROS 12, actuator, RTK-GPS	Research-grade platform
Tace et al. [10]	ML irrigation	KNN, SVM, NN, Node-RED, MongoDB	No autonomous rover
Morchid et al. [9]	Smart irrigation	ESP32, DHT22, moisture, water-level sensors	Stationary design
Saha et al. [26]	Precision agriculture	Satellite data, ML, fuzzy irrigation, IoT	Broader and more complex
Klaina et al. [37]	Communication	LoRaWAN, Sigfox, NB-IoT, Zigbee	No irrigation rover
KrishiRover	Autonomous monitoring	ESP32, 3 ultrasonic sensors, soil sensor, DHT11, pump, IoT, Bangla voice	Field validation still limited

III. METHODOLOGY

The KrishiRover was developed using the iterative prototyping technique in which each of the system's components including the structure, sensors, motors, irrigation equipment, communication systems, dashboard, and voice recognition systems were tested individually before the final assembly. It made it much easier to detect errors within each of the modules without affecting the rover as a whole. After successful completion of the tests, all the individual modules were combined into one complete agricultural rover. It is equipped with a four-wheel drive chassis and L298N motor driver allowing controlling the motors on both sides independently. The heart of the project is ESP32 microcontroller (MCU) which would coordinate processing information from the sensors and controlling drive, sprinkling, Wi-Fi, JSON,

and audio functions. The rover is equipped with three HC-SR04 ultrasonic distance sensors mounted on the front, left, and right parts of the chassis. To ensure the irrigation process is efficient, soil moisture sensor serves as an input whereas DHT11 senses temperature and humidity. The relay controls the pump. The operation of the soil and water pump is controlled by red and green LED indicators respectively. Sound is produced by the MAX98357A I2S amplifier.

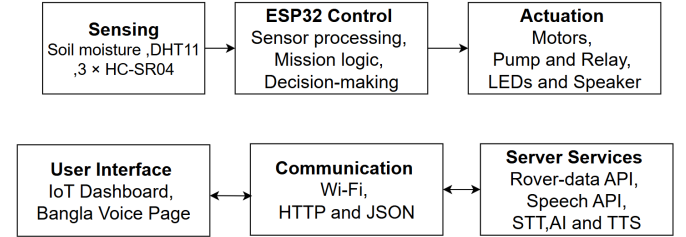


Fig. 2. Overall sensing, control, communication, and actuation architecture of KrishiRover.

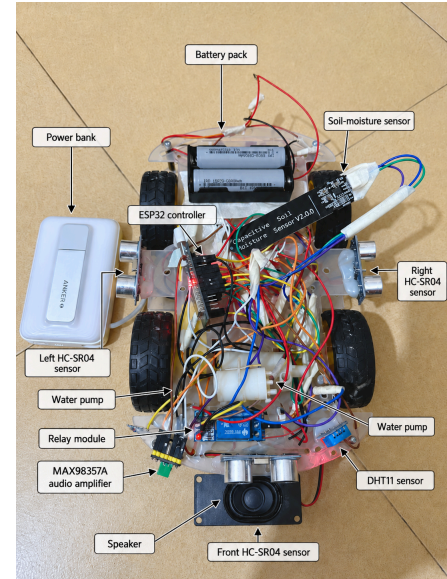


Fig. 3. Physical prototype of KrishiRover showing its main hardware components.

A C++ version compatible with Arduino was used to create the attached firmware. It uses Wi-Fi and HTTP transmission protocols for its communication, the ArduinoJson for the data transfer, the DHT sensor library for gathering environmental information, and the ESP32 audio libraries for I2S playback. The rover transmits to the local server its most recent readings of soil moisture, soil condition, temperature, humidity, pump status, distance to the obstacles, current zone, movement direction, and the current state of the rover. The system using the Bangla language uses the same information obtained from the rover. The user is able to ask a question using a web interface or a microphone, and the server replies in Bangla using the data from the rover. The system is designed in such a way that no information is generated based on current data, and when no information is available, the assistant is prevented from generating new sensor data.

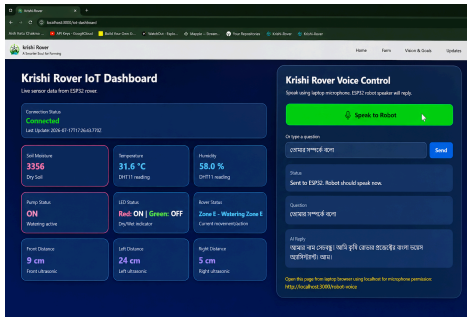


Fig. 4. KrishiRover IoT dashboard and Bangla voice-control interface used for live monitoring and human-robot interaction.

There are seven logical zones in which the route goes through from A to G. It starts from zone A and finally reaches G to return to A after going back in the reverse direction. At every zone, the rover stops for some time to collect data from the sensors to see if irrigation is required or not. The rover checks every zone for two seconds and if the readings are more than 2500 ADC then the relay-operated pump works for two seconds. If the readings are less than the threshold, the pump does not operate and the rover goes to the next zone.

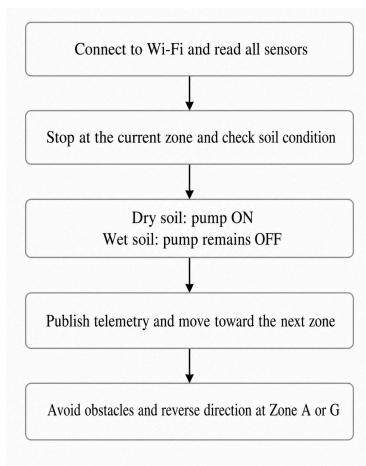


Fig. 5. KrishiRover IoT dashboard and Bangla voice-control interface used for live monitoring and human-robot interaction.

Obstacle avoidance relies on the use of a distance-based reactive controller for maneuvering the system. The front sensor defines the notice, warning, danger, and emergency ranges for the front, while the side sensors determine if the rover needs to change its movement heading to left or right. The current threshold distances for the front rotor are set to 28 cm, 20 cm, and 14 cm. The emergency distance is set to 8 cm. The side sensors are set at warning distance of 14 cm and danger distance of 9 cm, while the obstacle is considered to be in the emergency state when the distance is below 6 cm. When the obstacles get too close, the rover is halted, briefly reversed, and turned towards the side having more space available. The readings are ignored when the distances are above 180 cm or not comfortable to the controller. This is quite an efficient and lightweight method that does not require the use of a camera, LiDAR device, map, or powerful processor. Implementing more advanced navigation methods based on computer vision

or LiDAR, SLAM, or learning approaches is possible [27], [28]. The decision algorithm for the irrigation uses a clear threshold-based decision, which does not use machine learning techniques. Soil sensor generates analog data, and high values are considered as dry soil at the moment. Once the soil moisture reaches the dry level, ESP32 turns on the relay in active-low mode and activates the pump for some time interval. After the pump is switched off, the rover starts its motion once again, and watering stops in case the sensor stays above the threshold level. Since the ADC values are dependent on the type of the sensor and soil, the threshold needs to be calibrated through multiple readings of different soils – dry, average, and saturated. IoT data is published by the rover at an interval of about 2 seconds and it looks out for new voice requests from the server at an interval of 1.2 seconds. This ensures real-time monitoring without putting additional pressure on the ESP32 device and the network connection. All timings and distances are fully customizable to cater for various testing scenarios.

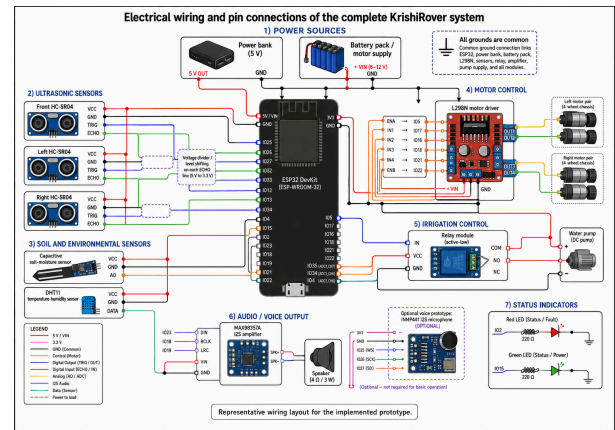


Fig. 6. KrishiRover IoT dashboard and Bangla voice-control interface used for live monitoring and human-robot interaction.

The full wiring integrates the ESP32 with the motor driver, ultrasonic sensors, soil moisture sensor, DHT11, relay, pump, indicator LEDs, audio amplifier, speaker, and regulated power supply. Power supply for the motor and pump is independent from that of low voltage controls to avoid interference. The overall wiring diagram indicates the GPIO pins in use, power specifications, common grounds, power distribution, and any resistive voltage protection between five volt sensors and ESP32. Functional tests were first performed at component level before the rover was fully assembled. The tests covered the direction of movement of the motors, the ultrasonics, soil characterization, readings from the DHT11 sensor, triggering the pump, telemetry, dashboard update as well as audio and responses in Bangla. The integrated testing subsequently determined if the rover could successfully move around the zones, steer clear of obstacles and stop in the respective zones, irrigate the dry soil and report its state during the audio playback session.

IV. SYSTEM DESIGN

KrishiRover was developed to be a complete agricultural platform that integrates field sensors, embedded control systems, mobility, irrigation, communication, and voice operations in Bangla. There are four sensors: beginning with the soil moisture sensor, then the DHT11, and lastly three ultrasonic sensors to detect the soil, surrounding environment, and barriers. Then the reading is received by the ESP32 to start controlling the motors, water pump, indicator LEDs, and speaker. The function of the system starts with sensing. The soil moisture sensor checks soil moisture to inform whether irrigation is needed, while the DHT11 helps measure temperature by providing additional data. The HC-SR04 sensors enable detection of obstacles along the path.

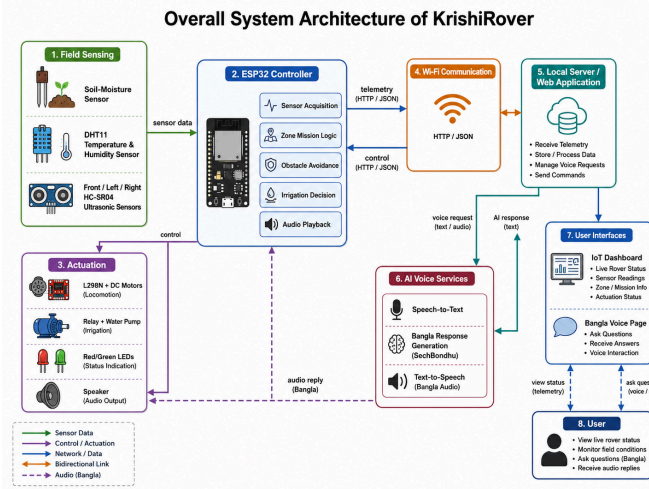


Fig. 7. Overall system architecture of KrishiRover showing the relationships among field sensing, embedded control, actuation, communication, server services, user interfaces, and Bangla voice interaction.

The system has two interfaces through which the user will input commands. The Internet of Things (IoT) dashboard displays real-time readings for soil, temperature, humidity, pump activity, distance from obstacles, zone where the rover is, direction of movement, and operation status. The Bangla voice interface takes either typed or spoken queries and produces brief answers based on the latest rover data. Both interfaces connect to the local server that handles voice requests and generates the telemetry data. AI voice services perform voice recognition, generate the answer using the most recent rover information, translate it into Bangla, and transform it into audio format. The audio is sent to the ESP32 chip for playing on the rover's loudspeaker. The software is organized into embedded, communication, server, web interface, and voice processing systems. The ESP32 firmware incorporates the functionalities for reading sensors, checking obstacle, zonation, activating irrigation, publishing telemetry data, and playing audio. The communication layer sends structured JSON messages between the rover and local server using Wi-Fi. This cooperation gives clarity to the software, as it simplifies testing and prevents the mixing of dashboard or voice processing operations with the rover's navigation

and irrigation logic. There are two primary software services provided by the back end. The rover-data service collects the telemetry data and stores the latest state of the rover. On the other hand, the speech-request service answers the queries sent by users via the Bangla voice interface. The dashboard queries the current state periodically and displays it for the user's benefit. When the user sends a voice or text query, the AI voice pipeline looks for the latest telemetry data to be included in the response. The system is instructed to generate no sensor reading in case telemetry data is not available.

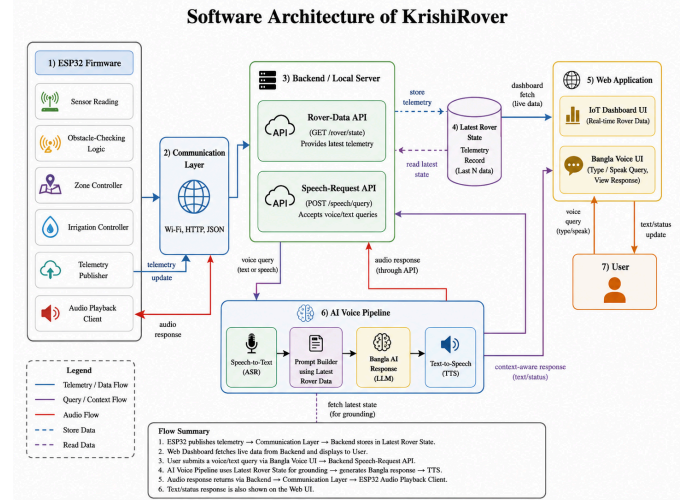


Fig. 8. Software architecture and data flow of KrishiRover, including the ESP32 firmware, communication layer, backend services, live rover state, web interfaces, and Bangla AI voice pipeline.

At all stages of the design process, safety and dependability were taken into account. Deterministic ESP32 programming governs the movement and irrigation process, as opposed to relying on uncontrolled audio commands. The pump is turned off and the motors and the pump stop when the Bangla audio is being played back. Both the dashboard and audio system adopt a common telemetry structure so that the numeric and audio displays are in agreement. The design allows the rover to maintain essential control functions on site while utilizing the server and web application for monitoring, explanation, and user interaction.

V. TEST DATA

This KrishiRover prototype was evaluated as an integrated unit to ensure the functioning of all these features together, namely its sensor system, irrigation system, obstacle detection system, IoT system, and communication in Bangla. In this recorded demonstration, the rover was connected to the local server and was functioning in zone E. The dashboard displayed a soil moisture reading of 3356 ADC, and since this was greater than the dry soil threshold set at 2500 ADC, the soil was considered dry, and the water pump started working. Also, in the same test, the temperature was 31.6 °C, relative humidity was 58.0%, and obstacle distances were 9 cm, 24 cm, and 5 cm at the front, left, and right sides respectively.

TABLE II
RECORDED DATA FROM THE KRISHIROVER PROTOTYPE DEMONSTRATION.

Measured item	Value	Interpretation	Observed response
Soil moisture	3356 ADC	Dry soil	Pump activated
Temperature	31.6 °C	Ambient condition	Displayed on dashboard
Relative humidity	58.0%	Ambient condition	Displayed on dashboard
Front distance	9 cm	Close obstacle	Distance reported
Left distance	24 cm	More open path	Distance reported
Right distance	5 cm	Very close obstacle	Distance reported
Current zone	Zone E	Active field zone	Watering Zone E
Pump state	ON	Irrigation active	Status displayed

Before performing watering or irrigation, the soil values were first compared with the predefined threshold. Since the obtained value was 3356 ADC, which was above the threshold of 2500 ADC, it was concluded by the system that the condition of the soil was dry leading to the activation of the pump using relays, switching on the indicator light and updating the rover with the task of irrigation in zone E. Thus, the test showed that all of its elements of soil sensing, decision-making, watering control, and reporting via IoT worked smoothly together as a single process. The testing of the Bangla interaction system was carried out using the phrase “তোমার সম্পর্কে বলো”. The interface accepted the input in Bangla, formed a response in Bangla mentioning SechBondhu’s identification as the KrishiRover assistant, and indicated that the answer was sent to ESP32 for speech reproduction. As the response timing and speech recognition accuracy were not evaluated, no numerical estimates on those two values have been provided.

TABLE III
FUNCTIONAL TEST RESULTS OF THE KRISHIROVER PROTOTYPE.

Test function	Result	Observation
Soil classification	Successful	3356 ADC identified as dry soil
Automatic irrigation	Successful	Pump activated in Zone E
Environmental sensing	Successful	Temperature and humidity received
Three-direction sensing	Successful	Front, left, and right distances received
IoT connection	Successful	Dashboard displayed a connected state
Dashboard consistency	Successful	Sensor and actuation values shown together
Bangla text query	Successful	Question accepted through the voice page
Bangla response	Successful	Relevant Bangla reply generated
ESP32 audio request	Successful	Interface confirmed that reply was sent

VI. RESULTS AND ANALYSIS

The prototype was successful in showing how the core features of KrishiRover work. ESP32 sensed soil data, temperature, humidity, ultrasonics data, followed the configured control rules, irrigated the soil when it was dry, and sent the latest status of the rover to the dashboard. All the information about field data, obstacle distance, pump status, zone, and system activities were shown on a single dashboard. The connection between sensing and irrigation can be seen in the recorded test. The reading of soil value is 3356 ADC which is more than the configured dry threshold, and the result is the pump turned ON. Simultaneously, the dashboard showed dry soil, the pump ON, the red LED status, and the message “Zone E – Watering Zone E.” Ultrasonic data revealed that

all the three sensors were in communication with the ESP32 board. It is found that left side has the maximum available distance of 24 cm, whereas the front side and right side have the values of 9 cm and 5 cm respectively. These readings will help us in having the information in which direction we need to go to avoid the obstacles. One single recording on the dashboard is not sufficient for calculating the accuracy or success of obstacle avoidance.



Fig. 9. KrishiRover operating in an outdoor vegetation area during prototype testing.

TABLE IV
SUMMARY OF THE DEMONSTRATED KRISHIROVER SUBSYSTEM RESULTS.

Subsystem	Status	Demonstrated behavior
ESP32 control	Working	Collected data and coordinated system actions
Soil sensing	Working	Detected a dry-soil value of 3356 ADC
Irrigation	Working	Activated the pump in Zone E
DHT11 sensing	Working	Reported 31.6 °C and 58.0% humidity
Ultrasonic sensing	Working	Reported three directional distances
IoT telemetry	Working	Transferred rover state to the dashboard
Bangla interaction	Working	Accepted a query and produced a Bangla reply
Audio delivery	Working	Reply request sent to the ESP32 speaker

Moreover, the IoT and Bangla voice operations worked correctly as well during the presentation. The dashboard displayed the current rover status, while the voice page answered to the question in the Bangla language. All these indicate the relationship between embedded control, communication with the server, web monitoring, and local language. The primary advantage of KrishiRover is the use of the combination of multiple robot-plant functions with relatively low-cost and common hardware devices. In contrast to scientific mobile sensing units, it does not create georeferenced soil maps and does not employ RTK-GPS, robotic arms, and machine vision [15], [16], [17]. However, it enhances stationary monitoring and irrigation of IoT devices [9], [12], [38]. However, there are still some limitations with respect to the prototype. Ultrasonic sensors give varying results with tilted obstacles, slender obstacles, absorptive obstacles, and soft obstacles. One low-cost soil sensor might not cover the entire root zone, and the output of this sensor will depend on soil characteristics, insertion depth, and calibration. The time-based zone movement may depend on the wheel slip, battery status, and irregularities of the terrain. Wi-Fi imposes restrictions on the range of operation of the prototype, and Bangla language processing is contingent upon the availability of a local server.

VII. DISCUSSION

The KrishiRover demonstrates how various smart agriculture features can be integrated into a single affordable mobile platform. The prototype combines soil monitoring, environmental sensor technology, obstacle avoidance, autonomous motion, area watering, Internet of Things connectivity, and Bangla voice control. During testing, all measurements, including soil composition, pump status, light indicators, zone data, and dashboard readings, were consistent with one another. This suggests that all major hardware and software components function together as planned. The project also demonstrates practical effectiveness of simple control approaches. Irrigation based on certain threshold values is simple to learn, test, and operate, while three ultrasonic sensors allow achieving basic navigation without expensive cameras and LiDAR devices. Moreover, the Bangla voice function provides a valuable opportunity to interact with the rover using local language. Nonetheless, this prototype relies on cheap sensors and makes use of time-based zone movement, Wi-Fi connection and a local server responsible for voice processing. Thus, the system has to be regarded more as an educational and experimental farming rover than a fully-functional commercial product.

VIII. CONCLUSION

KrishiRover has been successfully built as an innovative small-scale agricultural rover, which is capable of soil monitoring on zones basis, irrigation, obstacle avoidance, IoT reporting, and Human Robot Interaction in Bengali language. The project proved that the sensor values can control both the actions and water flow, while the dashboard and speech interaction make it easier to monitor the rover. As a result, the described project serves as an effective example of how cheap components can be utilized in order to develop an integrated smart farming prototype. Even though more calibration and field tests are necessary, KrishiRover gives a good start for the future research in this area.

IX. FUTURE WORK

Future developments in KrishiRover would help to improve soil monitoring through calibration of the soil moisture sensor based on soil types and using several probes. Navigation could be enhanced by using wheel encoders, GPS, RTK-GNSS, cameras, LiDAR, or SLAM-based maps. Furthermore, long-range radio networks such as LoRa, LoRaWAN, or even cellular communication would enable the operating range of KrishiRover to go beyond the present Wi-Fi setting. The existing irrigation system of the rover can be enhanced through measuring the water output rate and then calculating the required watering time according to the plant type and other data. Later on, machine learning or fuzzy control approaches might be applied to estimate irrigation requirements on the basis of previous data on the soil and weather. The Bangla voice system would also allow offline speech recognition, natural dialogue with the user, and answering questions related to other aspects of agriculture. Finally, future tests of the robot

should be carried out outdoors in different soils, weather, battery, and field conditions.

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